Table 1: Fully observable NeuronBench tasks. Each task is a noiseless regression from the external current I_{ext}, membrane voltage V, and every channel's open fraction ϕ_c to the membrane vector field $\dot{V}$. All six worlds share the Hodgkin–Huxley sodium, potassium, and leak currents, $\dot{V}_{\text{HH}} = I_{\text{ext}} - 120\,\phi_{\text{Na}}(V - 50) - 36\,\phi_{\text{K}}(V + 77) - 0.3\,(V + 54.4)$, with $\phi_{\text{Na}} = m_{\text{Na}}^3 h_{\text{Na}}$ and $\phi_{\text{K}} = n_{\text{K}}^4$, and differ in one added channel (or, for na_fatigue, a modified ϕ_{Na}). Units are standard Hodgkin–Huxley (mV, ms, $\mu\text{A}\,\text{cm}^{-2}$, $\text{mS}\,\text{cm}^{-2}$, $C_m = 1\,\mu\text{F}\,\text{cm}^{-2}$). Inputs are sampled by scrambled Sobol sequences over $I_{\text{ext}} \in [-40, 20]$, $V \in [-95, 60]$, and $\phi_c \in [0, 1]$. z_rebound is used for meta-training; the other five worlds are held out.

Task	Ground-truth equation	Description
z_rebound	$\dot{V} = \dot{V}_{\text{HH}} - 4\,\phi_{\text{Z}}(V - 120)$	Unidentified low-threshold inward current Z, $\phi_{\text{Z}} = m_{\text{Z}}^2 h_{\text{Z}}$; de-inactivated by hyperpolarization, producing depolarization block.
h_sag	$\dot{V} = \dot{V}_{\text{HH}} - 5\,\phi_{\text{H}}(V + 30)$	Hyperpolarization-activated, non-inactivating inward current I_h, $\phi_{\text{H}} = m_{\text{H}}$; produces voltage sag and post-inhibitory rebound.
na_fatigue	$\dot{V} = \dot{V}_{\text{HH}}$	No added current; $\phi_{\text{Na}} = m_{\text{Na}}^3 h_{\text{Na}} s_{\text{Na}}$ includes a slow inactivation gate, producing use-dependent spike-count run-down.
ca_rebound	$\dot{V} = \dot{V}_{\text{HH}} - 3.2\,\phi_{\text{T}}(V - 120)$	Fast low-threshold T-type Ca^{2+} current, $\phi_{\text{T}} = m_{\text{T}}^2 h_{\text{T}}$; de-inactivated by hyperpolarization, producing a rebound burst on release.
d_type	$\dot{V} = \dot{V}_{\text{HH}} - 9\,\phi_{\text{D}}(V + 77)$	D-type K^+ current, $\phi_{\text{D}} = m_{\text{D}} h_{\text{D}}$; de-inactivated by hyperpolarization, producing delayed or suppressed firing.
textbook_M	$\dot{V} = \dot{V}_{\text{HH}} - 2.5\,\phi_{\text{M}}(V + 77)$	Textbook slow, non-inactivating M-type K^+ current, $\phi_{\text{M}} = m_{\text{M}}$; produces spike-frequency adaptation.